Universidad de San Andrés
Argentina

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Editors
Journal of Statistical Software

Dear Editors,

Thank you for granting permission to resubmit a substantially revised version of our manuscript. We took the editorial assessment as a request for a software and inferential redesign, rather than a request for cosmetic changes.

The revised manuscript is entitled "BrainNetTest: Global and Edge-Wise Inference for Binary Brain Network Populations in R." The package now uses validated S3 input and result classes, standard print, summary, plot, and data-frame methods, and a deliberately smaller public API. Global randomization inference is separated from multiplicity-controlled edge inference. The prior adaptively stopped "critical edge" procedure is no longer presented as inferential; its retained prefix-sum path is explicitly descriptive and its computational gain is benchmarked.

We also rebuilt the replication materials. The standalone `code-full.R` script regenerates every reported simulation and benchmark result, table, figure, and numeric macro, while a separate short `code.R` and `code.html` exercise a reduced study. Both scripts use only the installed package and relative paths. The included README maps all outputs, and the full run records session and provenance information.

The manuscript is now organized around the clinical brain-network problem, statistical method, software contract, classes, workflows, real-data application, validation, and computational performance. The ABIDE case study compares autism and control resting-state connectomes in both the full cohort and a sex-, age-, and motion-balanced sensitivity sample. In both samples the observed global statistic is extreme under label randomization, while no individual connection survives strict multiplicity control. The manuscript therefore reports a preprocessing-conditional population association rather than a causal, diagnostic, or biomarker claim. The standalone application script and derived data make this analysis reproducible without access to raw images.

A detailed point-by-point response accompanies this letter.

Sincerely,

Maximiliano Martino and Daniel Fraiman